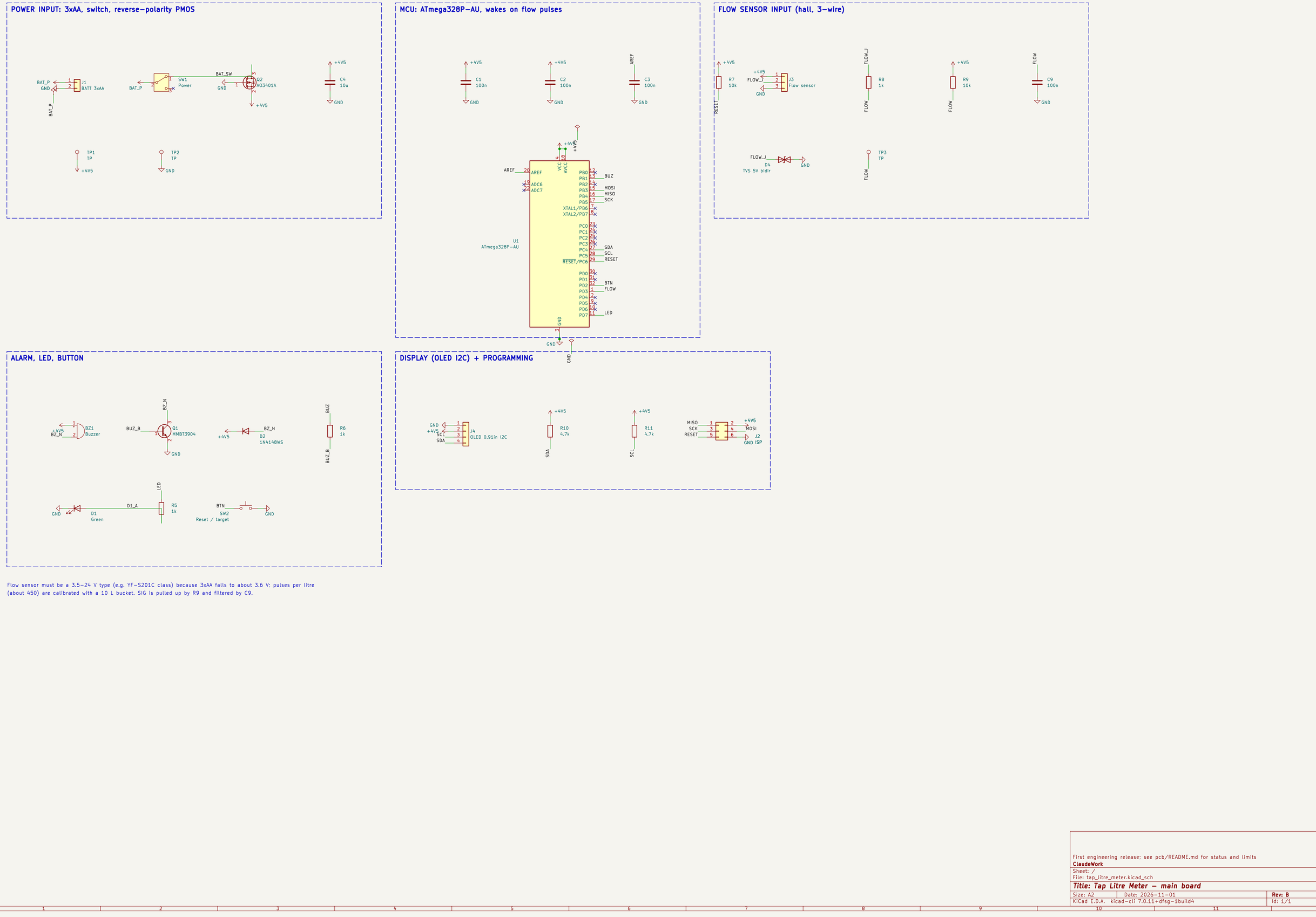

POWER INPUT: 3xAA, switch, reverse-polarity PMOS
BAT_P
GND
J1
BATT 3xAA
SW1
Power
BAT_SW
Q2
AO3401A
C4
10u
+4V5
TP1
TP2
MCU: ATmega328P-AU, wakes on flow pulses
C1
100n
C2
C3
AREF
U1
ATmega328P-AU
VCC
ADC6
ADC7
XTAL1/PB6
XTAL2/PB7
RESET/PC6
BUZ
MOSI
MISO
SCK
SDA
SCL
RESET
BTN
FLOW
LED
FLOW SENSOR INPUT (hall, 3-wire)
R7
10k
J3
Flow sensor
FLOW_J
R8
1k
R9
10k
C9
100n
D4
TVS 5V bidir
TP3
ALARM, LED, BUTTON
BZ1
Buzzer
BZ_N
BUZ_B
Q1
MMBT3904
D2
1N4148WS
R6
1k
D1
Green
R5
1k
SW2
Reset / target
DISPLAY (OLED I2C) + PROGRAMMING
J4
OLED 0.91in I2C
R10
4.7k
R11
4.7k
J2
GND ISP
MISO
SCK
MOSI
Flow sensor must be a 3.5–24 V type (e.g. YF-S201C class) because 3xAA falls to about 3.6 V; pulses per litre (about 450) are calibrated with a 10 L bucket. SIG is pulled up by R9 and filtered by C9.
First engineering release; see pcb/README.md for status and limits
ClaudeWork
Sheet: /
File: tap_litre_meter.kicad_sch
Title: Tap Litre Meter — main board
Size: A2
Date: 2026-11-01
Rev: B
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/1